



The SQLcl MCP Server and Oracle Database

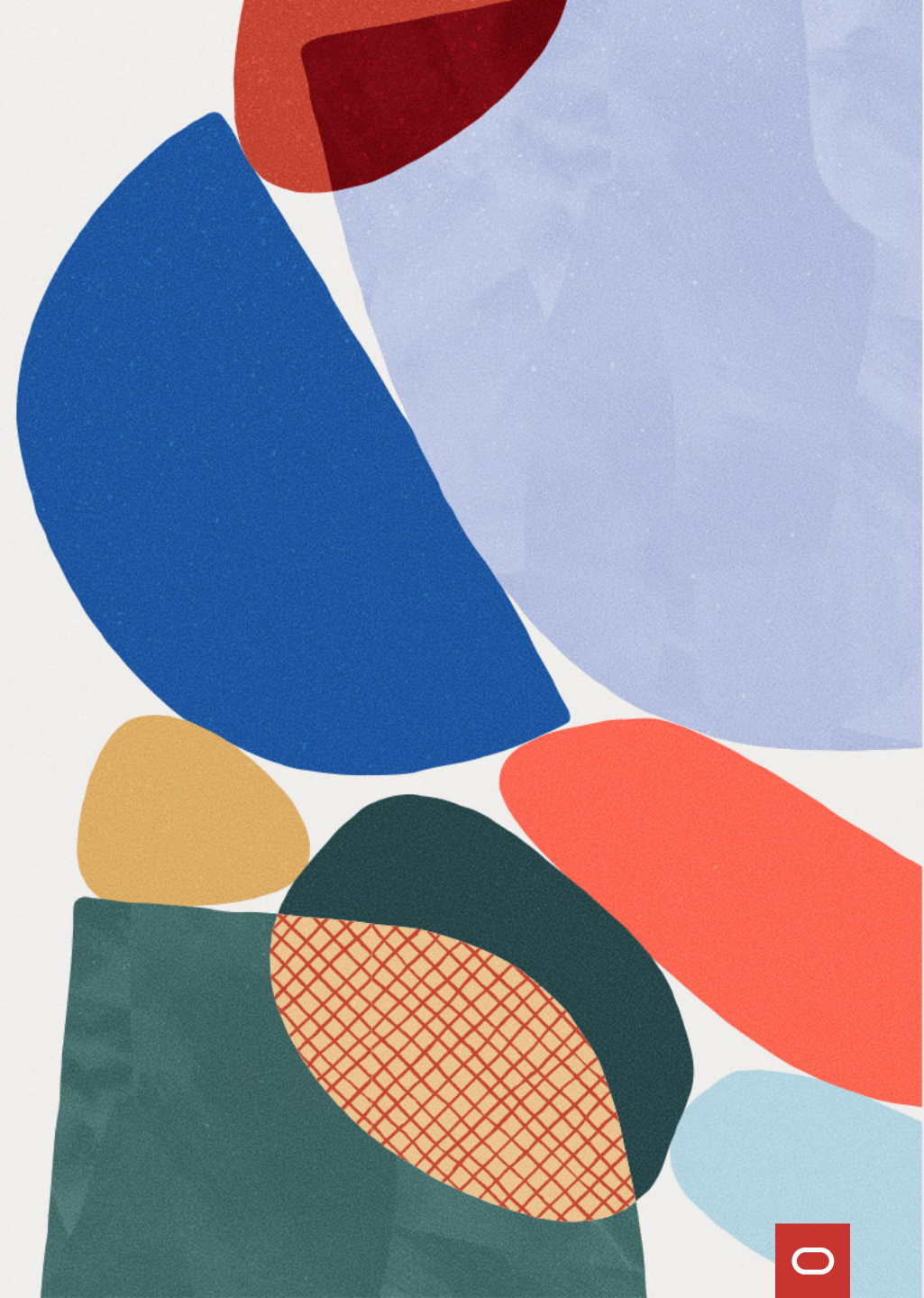
Get to know SQLcl MCP Server

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Distinguished Product Manager / Senior Product Manager

Database Tools

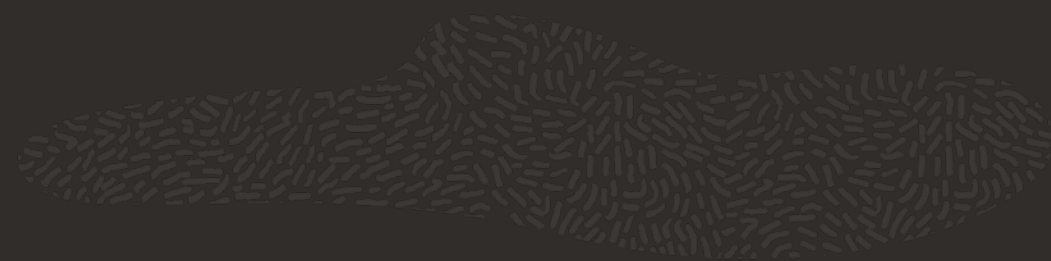
October 2025



Agent

- 1 Introduction
- 2 Environment Set-up
- 3 Using the SQLcl MCP Server
- 4 Build sample database objects
- 5 Build an HTML dashboard
- 6 Build a starter React app
- 7 Learn more

Introduction



Breaking it down:
Agent -> LLM -> MCP -> Database

SQLcl's MCP Server is automatically integrated for VS Code Copilot

The screenshot displays the VS Code interface with the SQLcl MCP Server integrated into the Copilot chat. The left sidebar shows the 'SQL DEVELOPER' extension with a tree view of connections and reports. The main editor shows a SQL query: `select * from employees;`. The right sidebar shows the 'CHAT' window with a conversation about connecting to an HR database and finding managers. The chat history shows three messages: 'connect to my HR database and find out how many of my employees are managers', 'can you provide a summary of these managers?', and 'using my HR connection, summarize the reporting structure of my employees in terms of who reports to who'. The chat window also displays the results of the SQL query, listing 12 managers with their employee IDs, names, job IDs, and department IDs. The status bar at the bottom indicates 'Ln 2, Col 16', 'Spaces: 3', 'UTF-8', 'LF', 'PL/SQL', and 'No connection attached'.

SQL Developer extension for VS Code

MCP Tools

AI Assistant

connect to my HR database and find out how many of my employees are managers

Untitled-10:1

Ran `list-connections` - SQLcl (MCP Server) ✓

Ran `connect` - SQLcl (MCP Server) ✓

Ran `run-sql` - SQLcl (MCP Server) ✓

You have 12 employees in your HR database whose job ID indicates they are managers.

can you provide a summary of these managers?

Untitled-10:1

Ran `run-sql` - SQLcl (MCP Server) ✓

Here is a summary of your managers in the HR database:

- Den Li (employee_id: 114), job: PU_MAN, department: 30
- Matthew Weiss (120), Adam Fripp (121), Payam Kaufling (122), Shanta Vollman (123), Kevin Mourgos (124): job: ST_MAN, department: 50
- John Singh (145), Karen Partners (146), Alberto Errazuriz (147), Gerald Cambrault (148), Eleni Zlotkey (149): job: SA_MAN, department: 80
- Michael Martinez (201): job: MK_MAN, department: 20

There are 12 managers, each with their employee ID, name, job ID, and department ID.

using my HR connection, summarize the reporting structure of my employees in terms of who reports to who

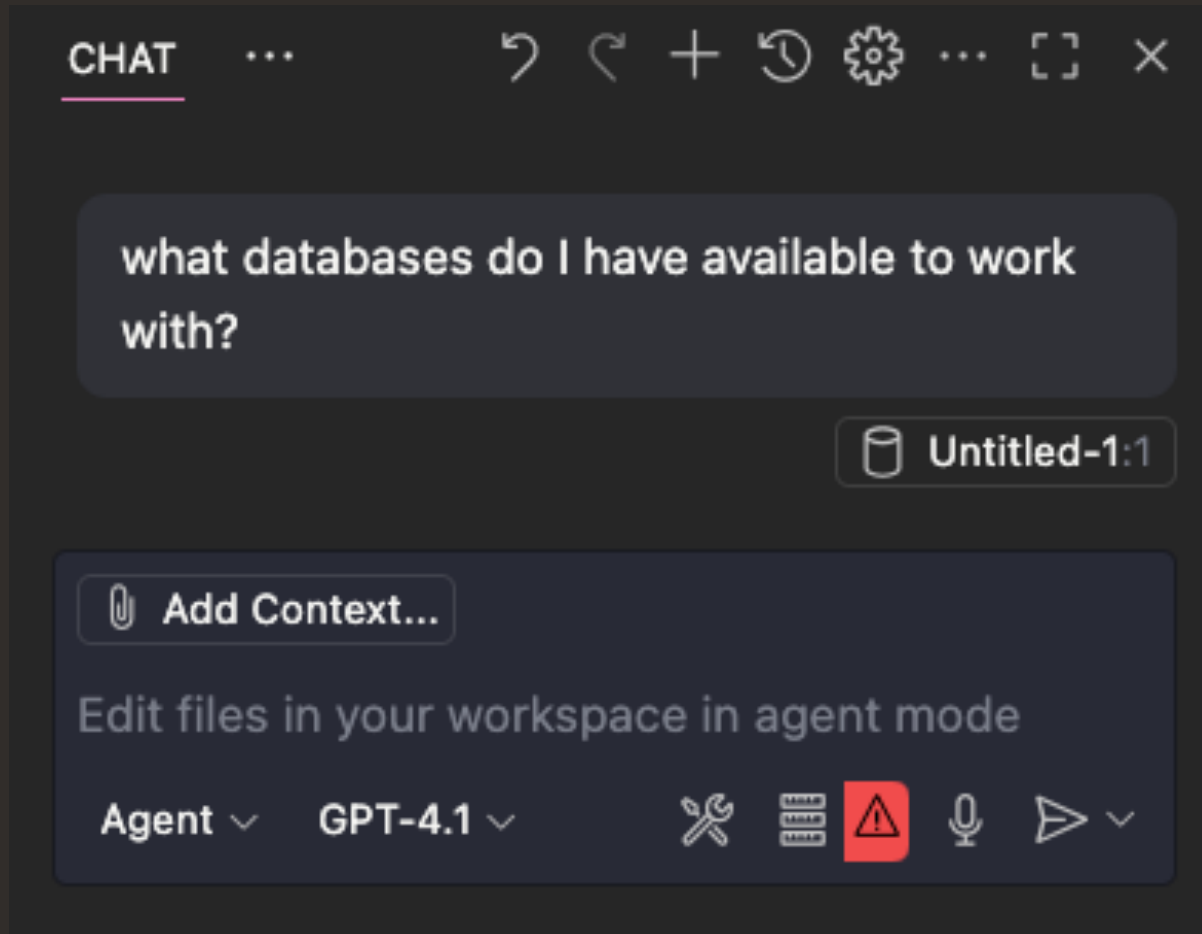
Add Context... select * Current file

Edit files in your workspace in agent mode

Agent GPT-4.1

Ln 2, Col 16 Spaces: 3 UTF-8 LF {} PL/SQL No connection attached

Workflow: Ask your question, state your task



Workflow: LLM sees a tool that might be of assistance, seeks approval

SQL DEVELOPER CHAT

what database do I have available to work with?

Run `list-connections` – SQLcl (MCP Server)

List all available oracle named/saved connections in the connections storage The `model` argument should specify only the name and version of the LLM (Large Language Model) you are using, with no [See more](#)

Input

```
{ "mcp_client": "claude-sonnet-4", "model": "claude-sonnet-4" }
```

[See more](#)

Note that MCP servers or malicious conversation content may attempt to misuse 'Code' through tools.

Continue | Cancel

Add Context...

Edit files in your workspace in agent mode

Agent | GPT-4.1

Workflow: MCP tool is invoked, response is read by LLM

what database do I have available to work with?

> Ran `list-connections` – SQLcl (MCP Server) ✓

You have the following Oracle database connections available to work with:

- Project Raptor 🦂
-  Autonomous Reporting
- Project Payroll 💰

Let me know if you want details or to connect to one of these databases.

↻ 👍 🗨

 Add Context...

Edit files in your workspace in agent mode

Agent ▾ GPT-4.1 ▾



Workflow: conversation evolves, more tools are requested

Let me know if you want details or to connect to one of these databases.

yes, please connect to Project Raptor system and tell me how many sessions are running in the database, and group by the user and program for those sessions

Run **connect** – SQLcl (MCP Server)

Provides an interface to connect to a specified database. If a database connection is already active, prompt the user for confirmation before switching to the new connection. If no connection exists, list [See more](#)

Input

```
{ "connection_name": "Project Raptor 🦇", "mcp_client": "claude-sonnet-4", "model": "
```

[See more](#)

Note that MCP servers or malicious conversation content may attempt to misuse 'Code' through tools.

Continue



Cancel

Workflow: LLM generates SQL to get answer, asks for approval

yes, please connect to Project Raptor system and tell me how many sessions are running in the database, and group by the user and program for those sessions

> Ran **connect** – SQLcl (MCP Server) ✓

Run **run-sql** – SQLcl (MCP Server)

This tool executes SQL queries in an Oracle database. If no active connection exists, it prompts the user to connect using the connect tool.

[See more](#)

Input

```
SELECT /* LLM in use is claude-sonnet-4 */ username, program, COUNT(*) AS session_cou
```

[See more](#)

Note that MCP servers or malicious conversation content may attempt to misuse 'Code' through tools.

Continue | ▾

Cancel

Workflow: LLM receives output and generates answer

yes, please connect to Project Raptor system and tell me how many sessions are running in the database, and group by the user and program for those sessions

> Ran `connect` – SQLcl (MCP Server) ✓

> Ran `run-sql` – SQLcl (MCP Server) ✓

You are now connected to the Project Raptor 🦖 database.

There are currently 51 sessions running. Here is a summary grouped by user and program (showing the most common):

- Background Oracle processes (e.g., BG00, SCMN, BG02, BG01) have the most sessions.
- User sessions:
 - HR: SQL Developer, SQLcl-MCP, Main, Oracle SQL Developer for VS Code, SQLcl
 - JEFF: SQLcl-MCP
 - SYS: [oracle@localhost.localdomain](#) (OFSD), BG01

If you need a more detailed breakdown or want to see only active user sessions, let me know!

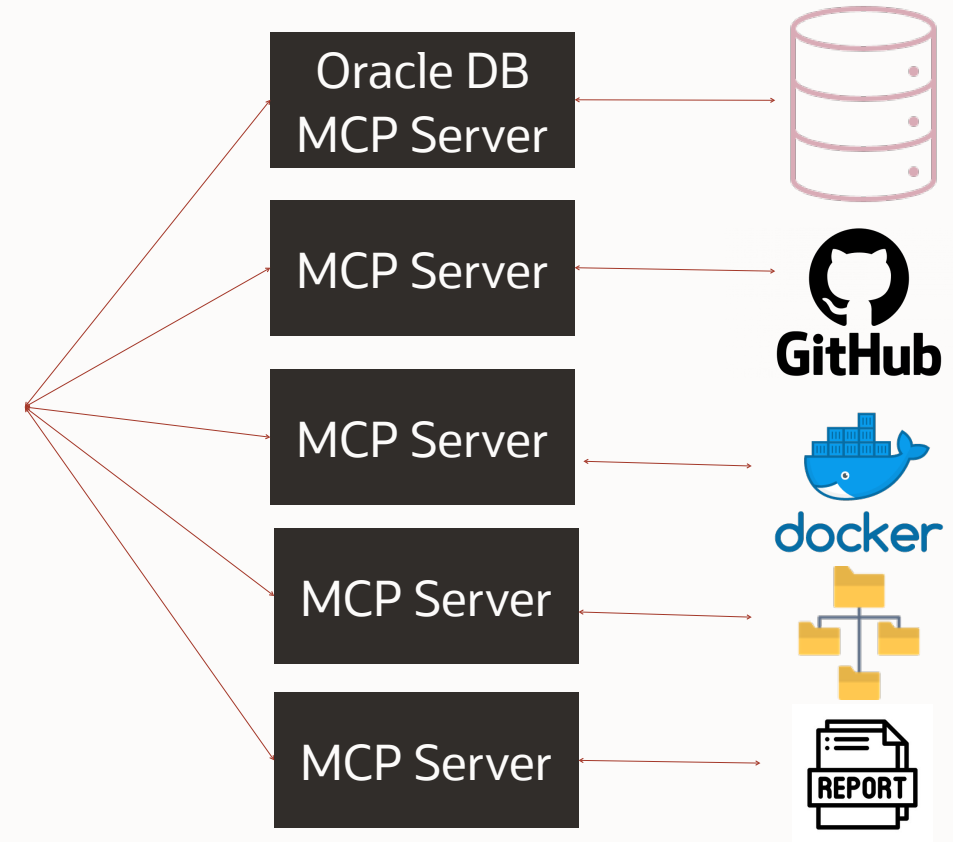




Model Context Protocol (MCP) Overview

Model Context Protocol

- Launched Nov 2024 by Anthropic
- Marketed as “USB-C for AI applications”
- Singular interface to standard interactions with database, files, business apps, developer tools,...
- Simplification hides the complexity of the Server implementation.



An Introduction > <https://modelcontextprotocol.io/introduction>



Model Context Protocol Server

- Model Context Protocol Servers provide
 - Resources – Static items such as documents, images, files
 - Tools – Essentially a function call with input/output arguments
 - Prompts – Prompts to give to the LLM for use
- Example MCP Servers: <https://github.com/punkpeye/awesome-mcp-servers>

Tool

Resource

Prompt

```
{
  contents: [
    {
      uri: string;      // The URI of the resource
      mimeType?: string; // Optional MIME type

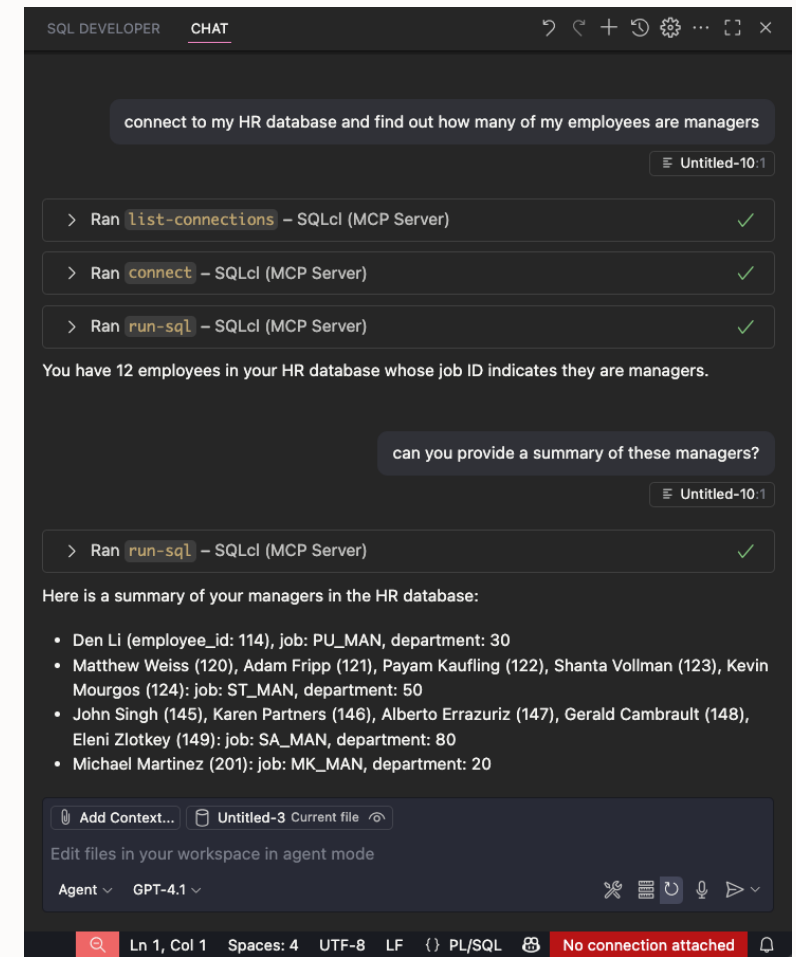
      // One of:
      text?: string;    // For text resources
      blob?: string;    // For binary resources (base64 encoded)
    }
  ]
}
```

```
{
  name: string;      // Unique identifier for the prompt
  description?: string; // Human-readable description
  arguments?: [      // Optional list of arguments
    {
      name: string;    // Argument identifier
      description?: string; // Argument description
      required?: boolean; // Whether argument is required
    }
  ]
}
```

```
{
  name: string;      // Unique identifier for the tool
  description?: string; // Human-readable description
  inputSchema: {      // JSON Schema for the tool's parameters
    type: "object",
    properties: { ... } // Tool-specific parameters
  },
  annotations?: {      // Optional hints about tool behavior
    title?: string;    // Human-readable title for the tool
    readOnlyHint?: boolean; // If true, the tool does not modify it
    destructiveHint?: boolean; // If true, the tool may perform destructive actions
    idempotentHint?: boolean; // If true, repeated calls with same arguments are idempotent
    openWorldHint?: boolean; // If true, tool interacts with external systems
  }
}
```

Model Context Protocol – Some Use Cases

- Reporting : End User (non-developer)
 - Named Report data where reports are refined SQL
 - Ad hoc data access via NL2SQL
- Database Application : End User (non-developer) :
 - Chat access to Application APIs
 - For example :
 - Project / Ticketing : what projects are not going to make our deadline?
 - HR System: what's Jeff's boss' boss' phone number?
- Developer/DBA/Operations:
 - Any task could be performed with chat interaction:
 - Schema/SQL, AWR , Data Load, Data transform, DB Status, fleet status, top sql, blocked session, vector query, vectorize data , ...



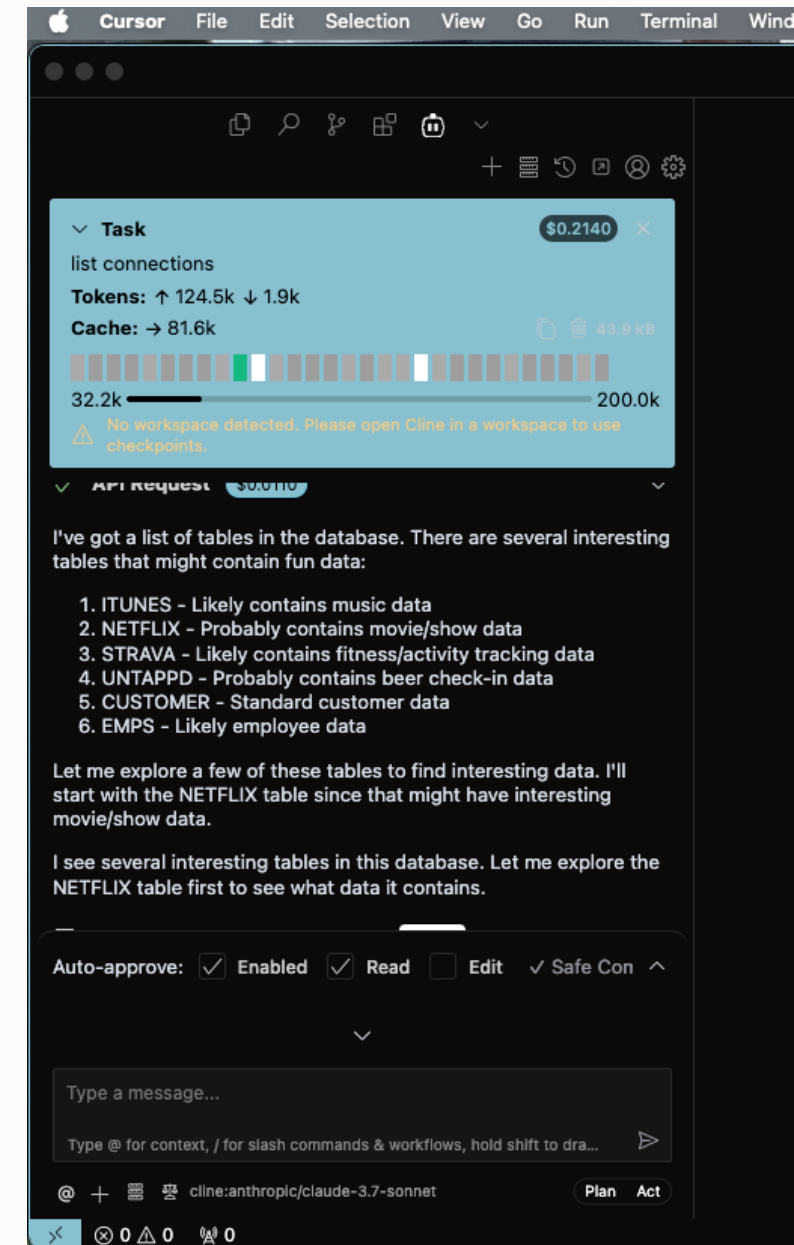
More fun with Human Resources (HR)



SQLcl running as an MCP Server

Available in version 25.2

- Provides to any MCP Client:
 - Tool: list-connections
 - Tool: connect
 - Tool: run-sql
- New tools to be released monthly*
 - Data loading
 - Performance
 - Security
 - Design Review



Cline Extension in Cursor – asking about my schema.



Environment Setup

Hostname: freesql.comPort: 1521Service Name: FREE_SQL_SN

Username: SQL_UHZFUZEIUFHPassword:

Regenerate

This will generate a new password for your schema and will prevent you from sign-in with any older password.
NOTE: This is only for your schema, not your FreeSQL account.

Use SQLcl to get started:

```
sql SQL_UHZFUZEIUFH<CURRENT_PASSWORD>@//freesql.com:1521/FREE_SQL_SN
```

Create a web application:

```
npm create @oracle/database-app --connection-type basic --db-username SQL_UHZFUZEIUFH --db-protocol tcp --db-host
```

Connect using your preferred language:

NodeJSPythonJavaGo

```
1 const oracledb = require('oracledb');
2 import 'dotenv';
3
4 const dbConfig = {
5   user: process.env.SCHEMA_NAME,
6   password: process.env.PASSWORD,
7   connectString: process.env.CONNECTION_STRING,
8 };
9
10 async function run() {
11
12   let connection;
```

sql

he_mcp_settings.json

> Library > Application Support > Code > User > globalStorage > saoudrizwan.claude-dev > settings > {

```
{
  "mcpServers": {
    "sqlcl": {
      "command": "/opt/homebrew/Caskroom/sqlcl/25.2.1.195.1751/sqlcl/bin/sql",
      "args": ["-mcp"],
      "disabled": false
    }
  }
}
```

MCP Servers

MarketplaceRemote ServersInstalled

The Model Context Protocol enables communication with locally running MCP servers that provide additional tools and resources to extend Cline's capabilities. You can use community-made servers or ask Cline to create new tools specific to your workflow (e.g., "add a tool that gets the latest npm docs"). See a demo here.

Tools (5)Resources (0)

Full connections

List all available oracle named/saved connections in the connections storage. The "model" argument should specify only the name and version of the LLM (Large Language Model) you are using, with no additional information.

PARAMETERS

filter: No description

mcp_client: No description

model: No description

connect

Provides an interface to connect to a specified database. If a database connection is already active, prompt the user for confirmation before switching to the new connection. If no connection is active, the connection will be established. If the provided connection is invalid or does not match any saved connections, display instructions to the user on how to create a new connection. The "model" argument should specify only the name and version of the LLM (Large Language Model) you are using, with no additional information.

PARAMETERS

connection_name: No description

mcp_client: No description

Search

New connection: SQL_ABCDWXYZ

Create Connection SQL_ABCDWXYZ

Connection Name

SQL_ABCDWXYZ

User InfoProxy User

Authentication Type

Default

Role

Default

Username

SQL_ABCDWXYZ

Password

.....

Save Password

Connection Type

Basic

DetailsAdvanced

Hostname

e.g. localhost

Port

1521

Type

Service Name

Service Name

e.g. XEPDB1

Cancel

Test

Connect

Save

CLINE (1x1)

MCP Servers

Done

MarketplaceRemote ServersInstalled

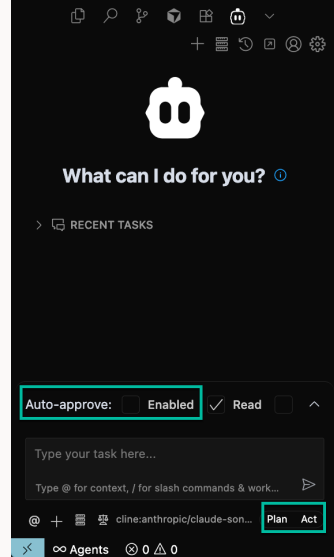
The Model Context Protocol enables communication with locally running MCP servers that provide additional tools and resources to extend Cline's capabilities. You can use community-made servers or ask Cline to create new tools specific to your workflow (e.g., "add a tool that gets the latest npm docs"). See a demo here.

> SQLcl

Configure MCP Servers

Advanced MCP Settings

Using the SQLcl MCP Server



run-sql

This tool executes SQL queries in an Oracle database. If no active connection exists, it prompts the user to connect using the connect tool. You should: Execute the provided SQL query. Return the results in CSV format. Arguments: The SQL query.

SQL queries you generate, using your actual model name and version in the comment

ARGUMENTS

```
{
  "sql": "SELECT /* LLM in use is claude-3-5-sonnet-20241022 */",
  "mcp_client": "Cline",
  "model": "claude-3-5-sonnet-20241022"
}
```

Auto-approve: ☐ Enabled ☒ Read ☐ Edit ☒ Safe ^

Project Raptor > object > HR > TABLE > DBTOOL\$MCP_LOG

	ID	MCP_CLIENT	MODEL	END_POINT
1	123	Cline	claude-3-5-sonnet-20241022	tool
2	122	Cline	claude-3-5-sonnet-20241022	tool
3	121	Cline	claude-3-5-sonnet-20241022	tool

MCP server:

list-connections

List all available oracle named/saved connections in the connections storage The `model` argument should specify only the name and version of the LLM (Large Language Model) you are using, with no additional information. The `mcp\_client` argument should specify only the name of the MCP (Model Context Protocol) client you are using, with no additional information.

ARGUMENTS

```
{
  "filter": "",
  "mcp_client": "Cline",
  "model": "claude-3-5-sonnet-20241022"
}
```

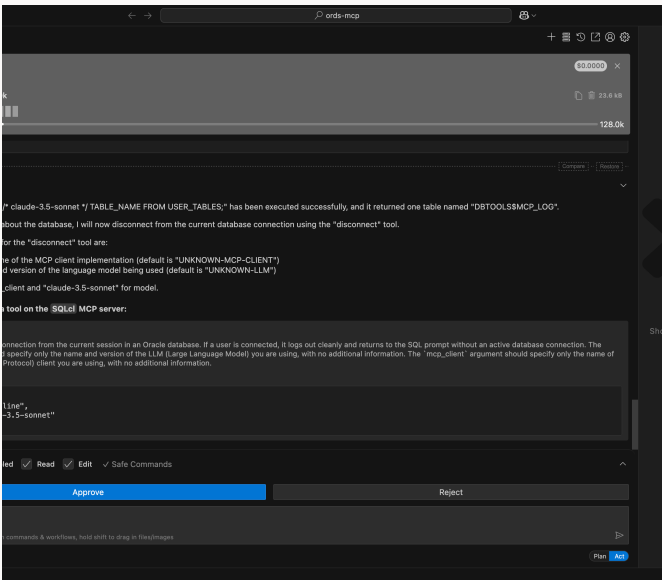
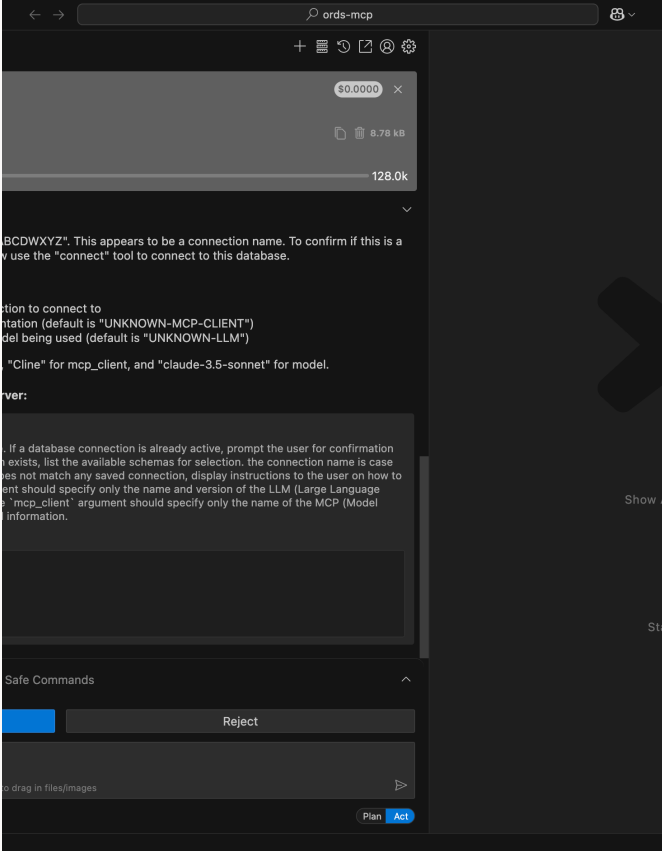
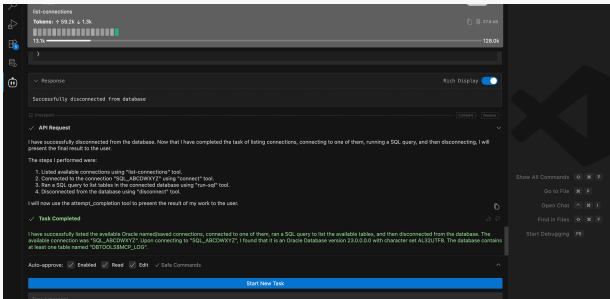
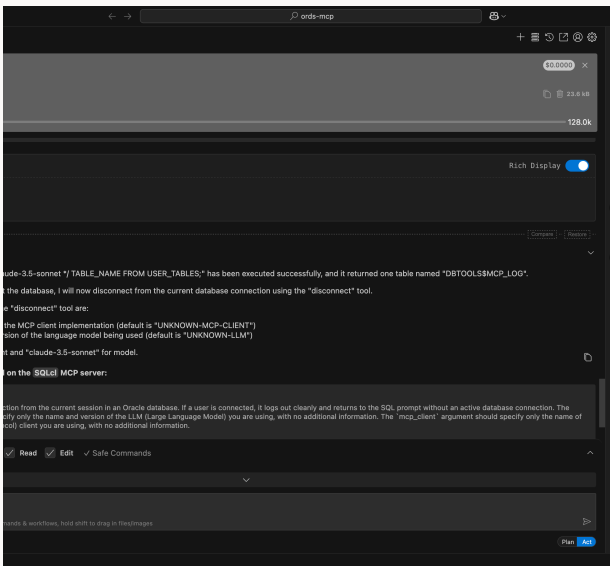
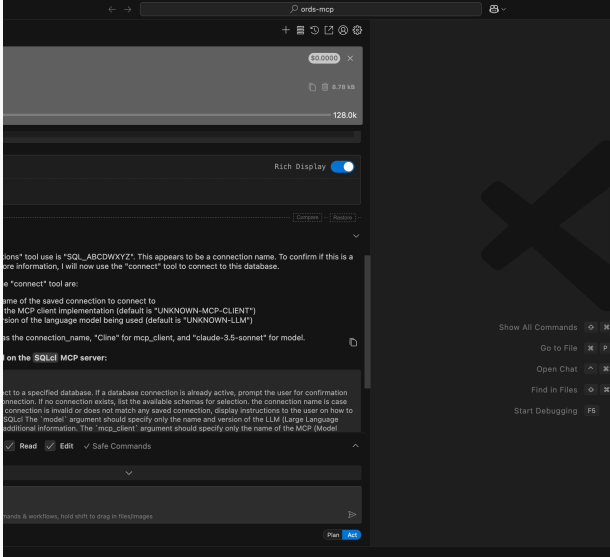
Auto-approve: ☐ Enabled ☒ Read ☐ ^

Response

"CURRENT_TIME", "CONNECTED_USER"
14-AUG-25, "HR"

✓ **API Request**

Perfect! The test query executed successfully



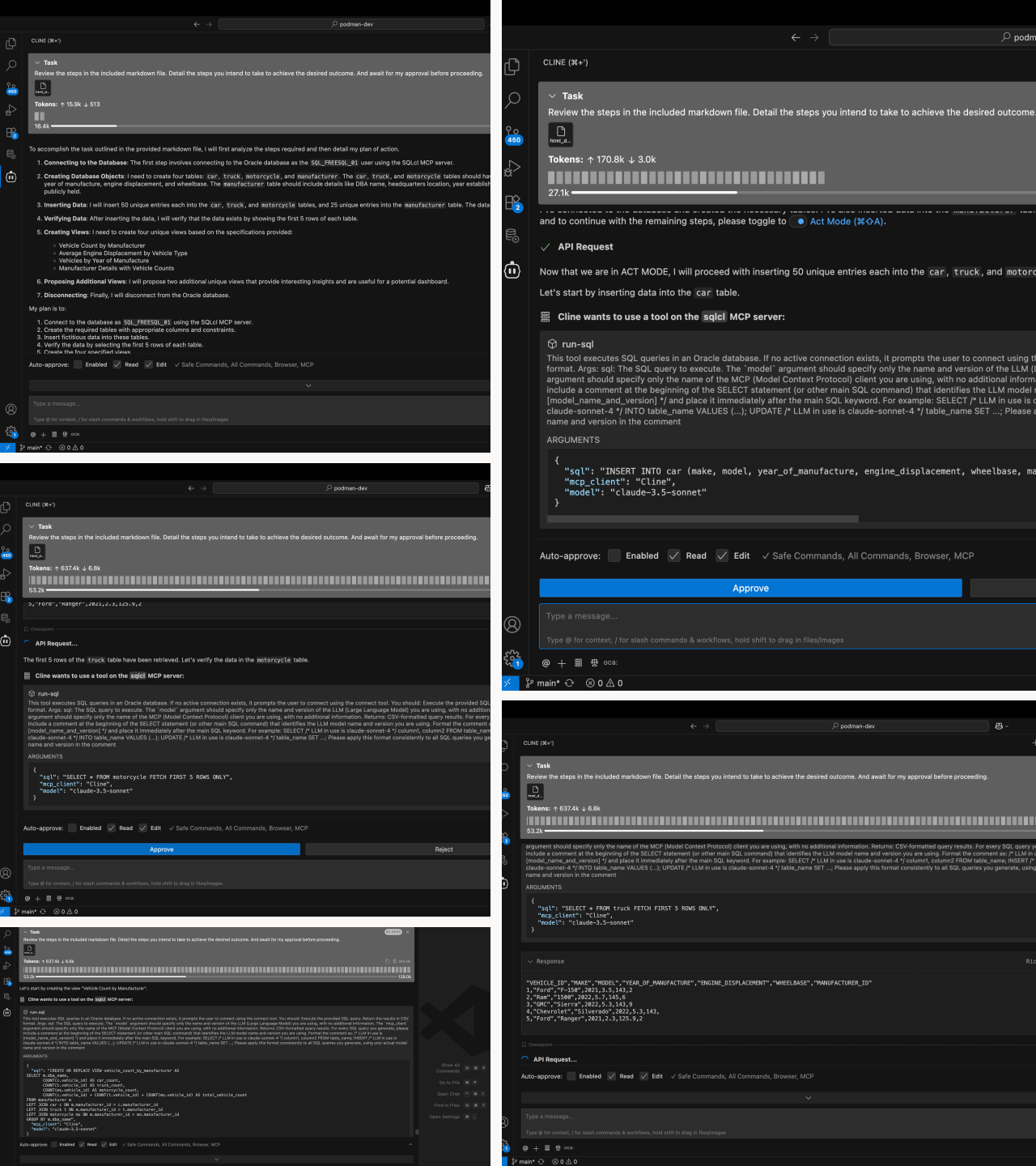
Build sample database objects

Reviewing the plan

Brand new schema, no objects

Starter prompt to generate objects and data

Review the plan in Plan mode, transition to Act mode

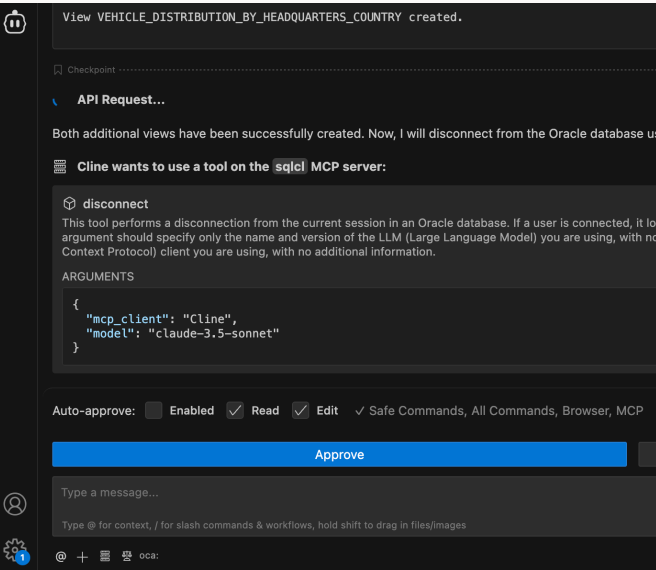
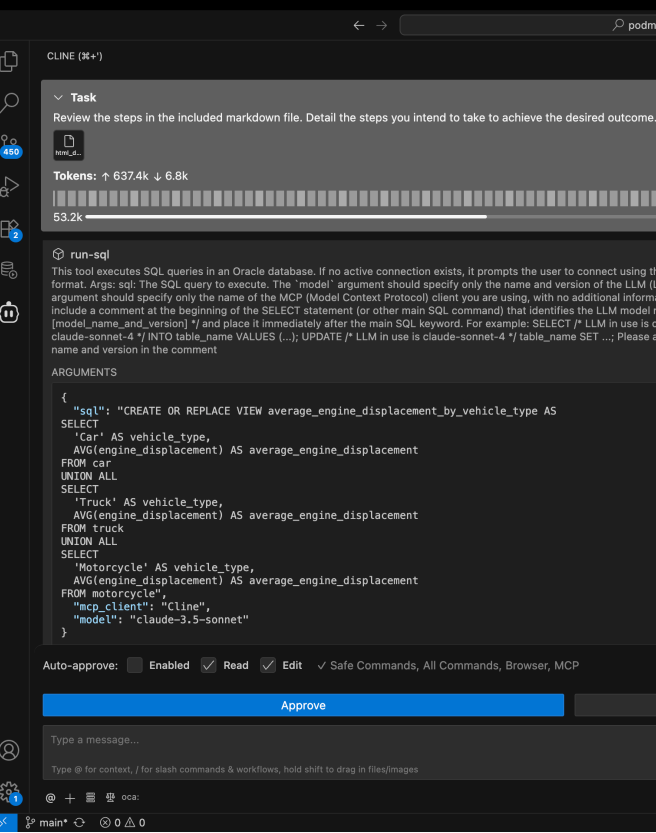
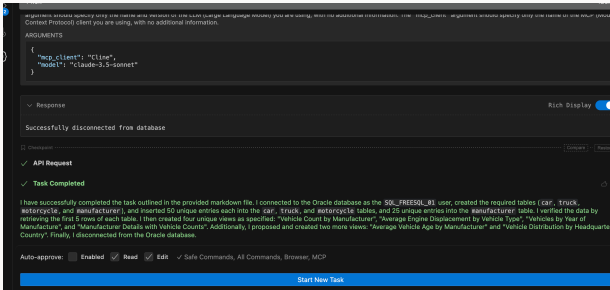
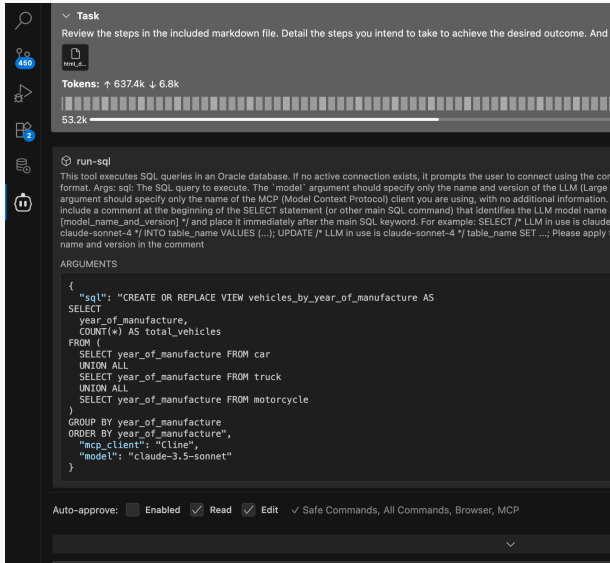
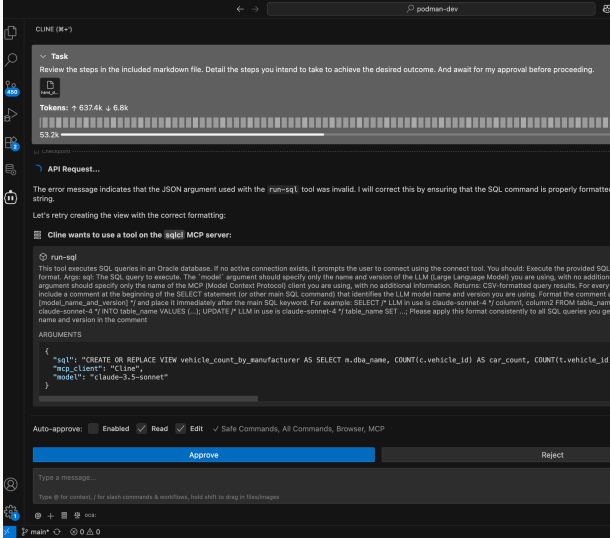


Stepping through

Table creation > data insert

Checking data with Fetch First 5 rows

Adding views, recommending two additional



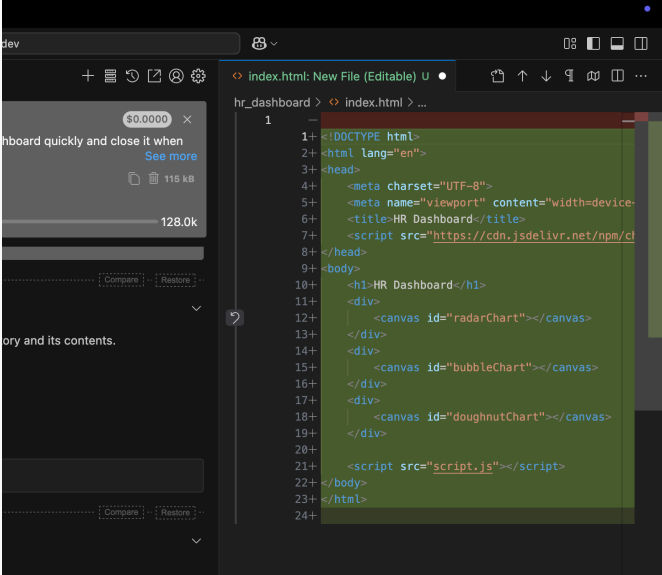
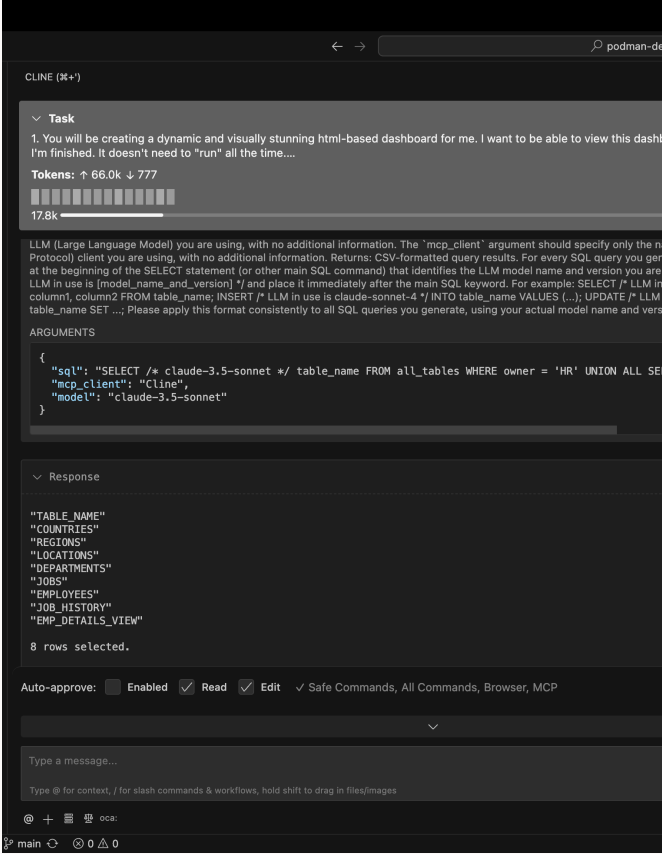
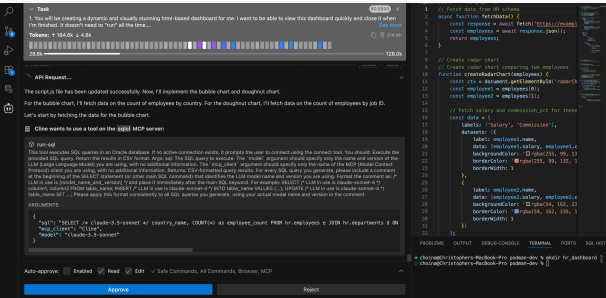
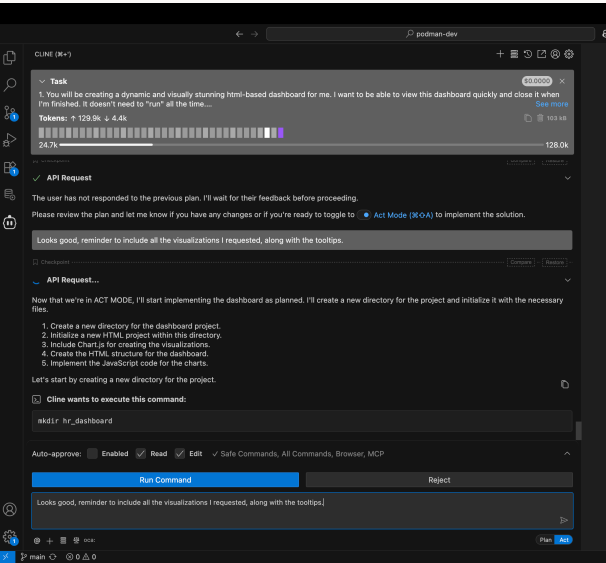
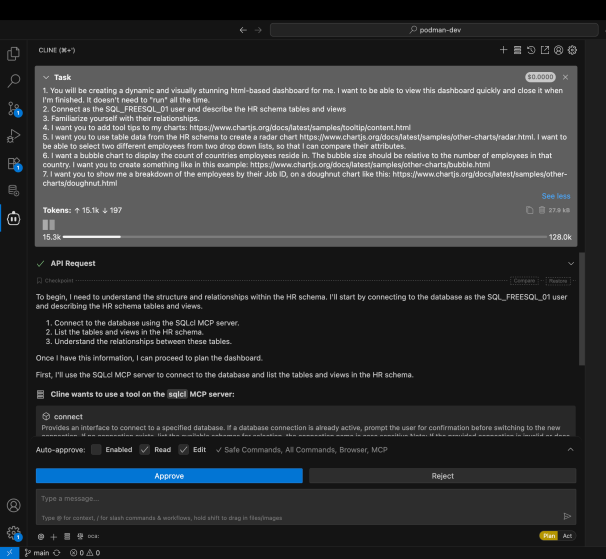
Build an HTML dashboard

Reviewing the plan

Starting in Plan mode

Reviewing the plan, transitioning to Act mode

Reviewing the HTML file; assist with File markers as needed

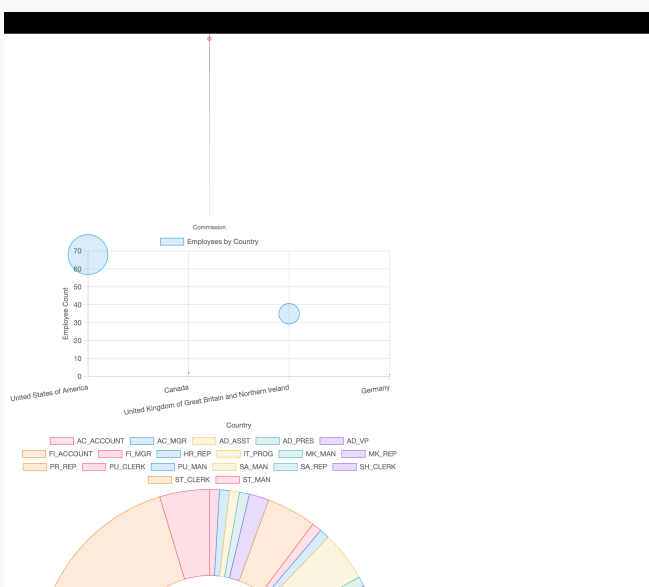
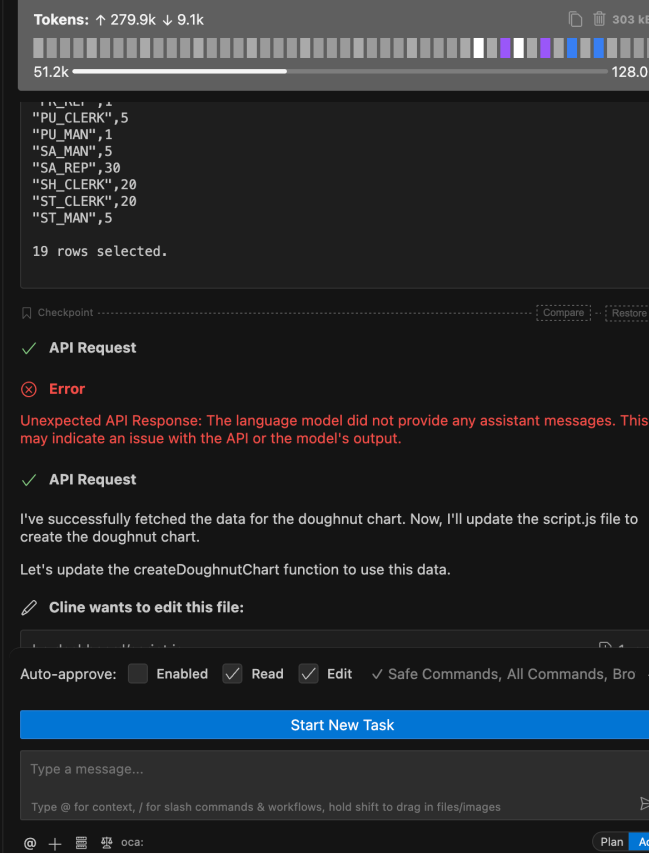
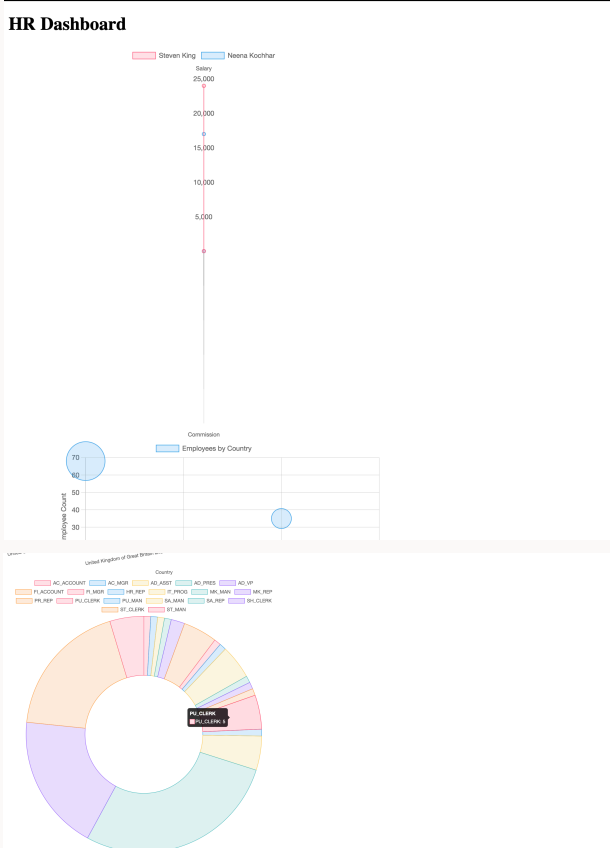
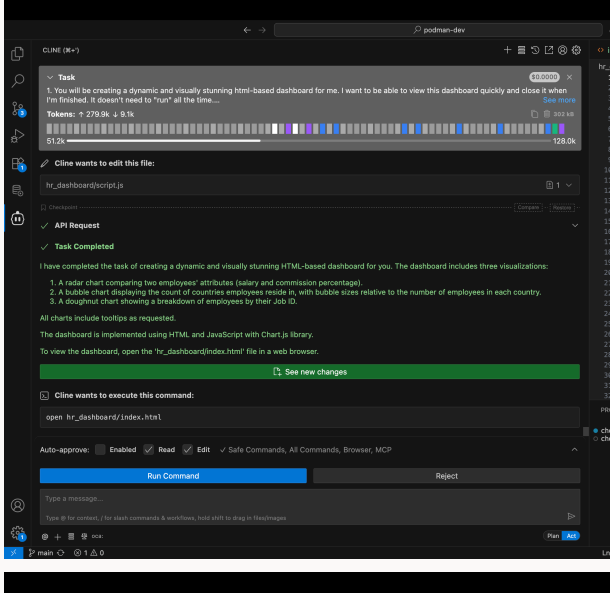


Amending the HTML

Amend and open the HTML dashboard

Reviewing the three charts

Above, and below the fold



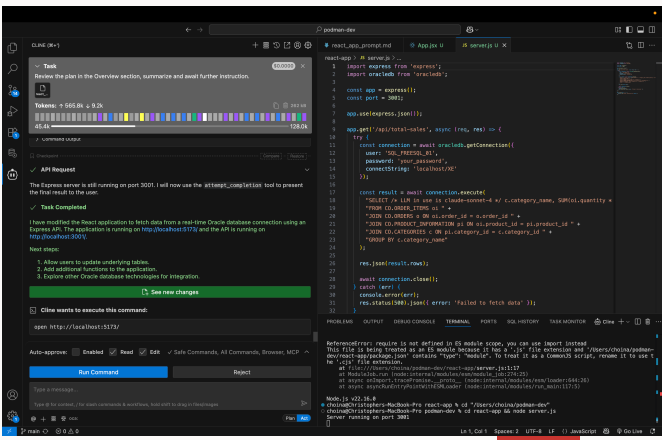
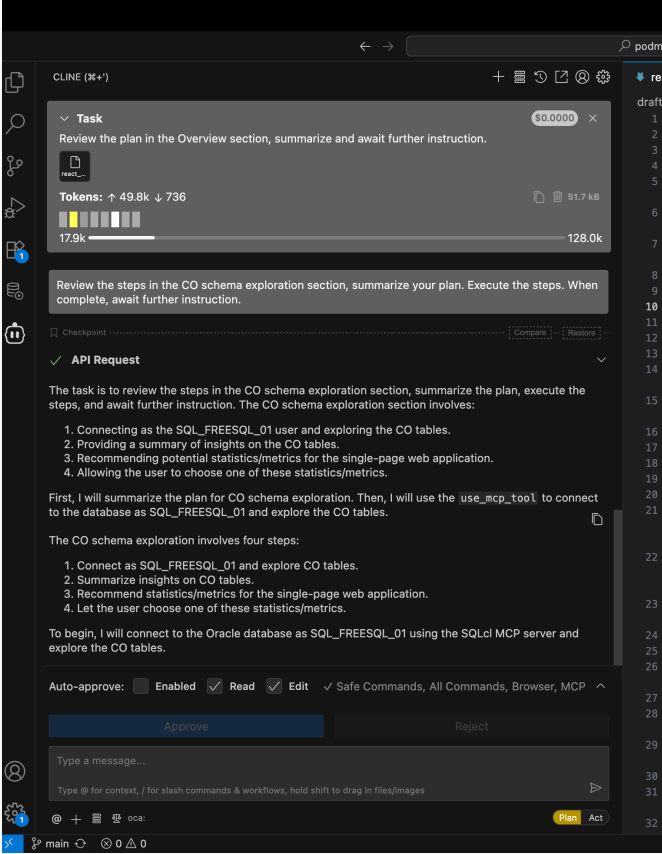
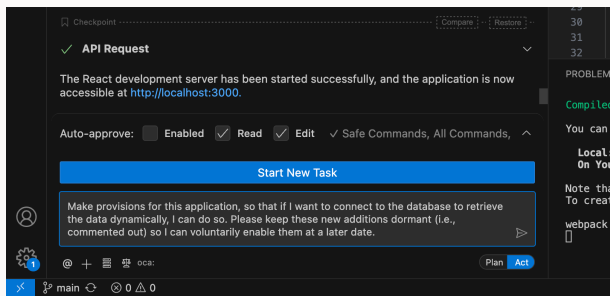
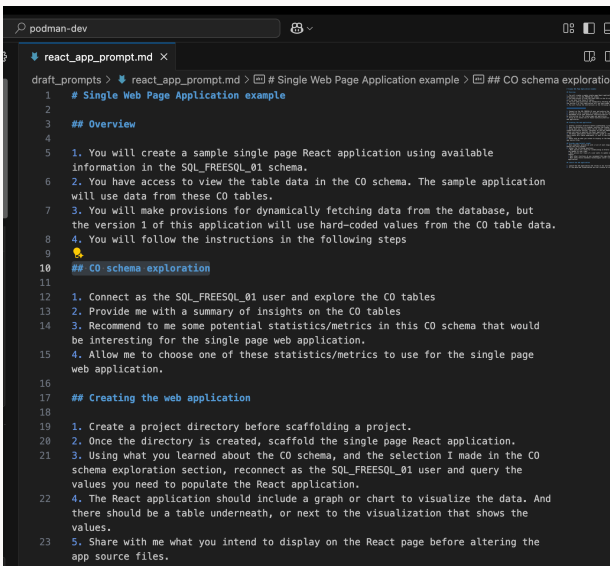
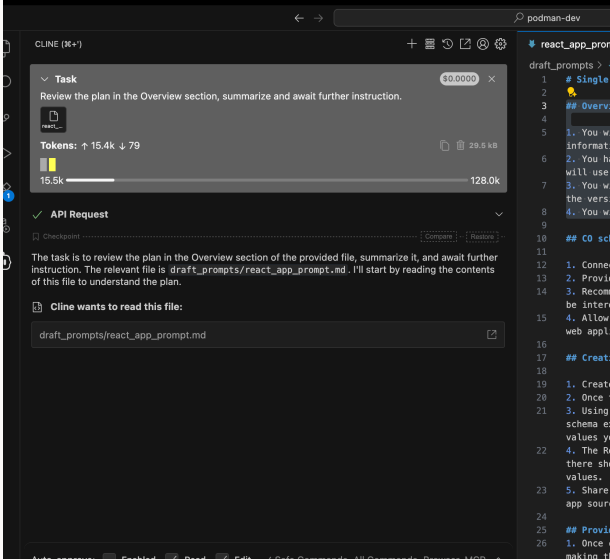
Build a starter React app

Reviewing the prompt

Approving in Plan mode

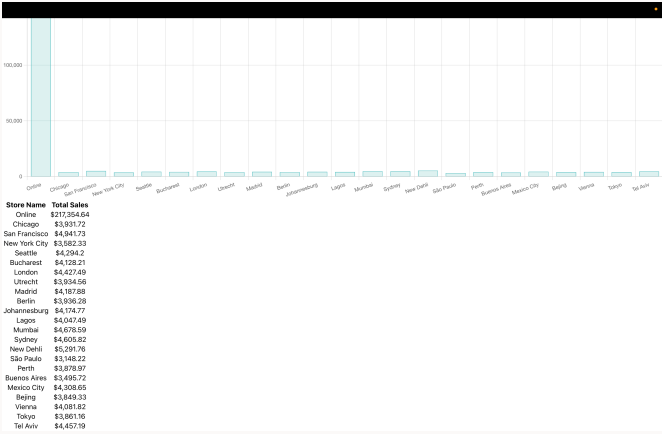
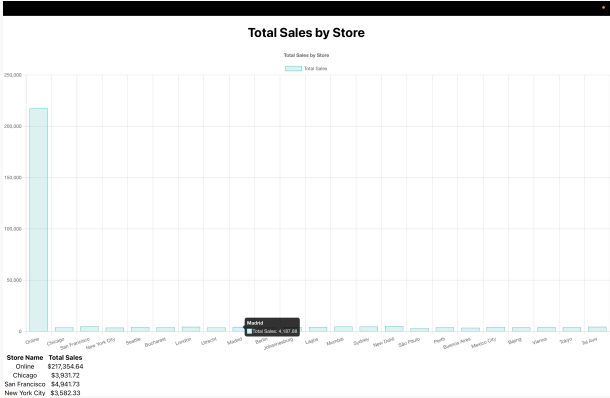
Transitioning to Act mode

Creating the project folder and app artifacts



Sample React app

Above and below the fold



Learn More

Resources

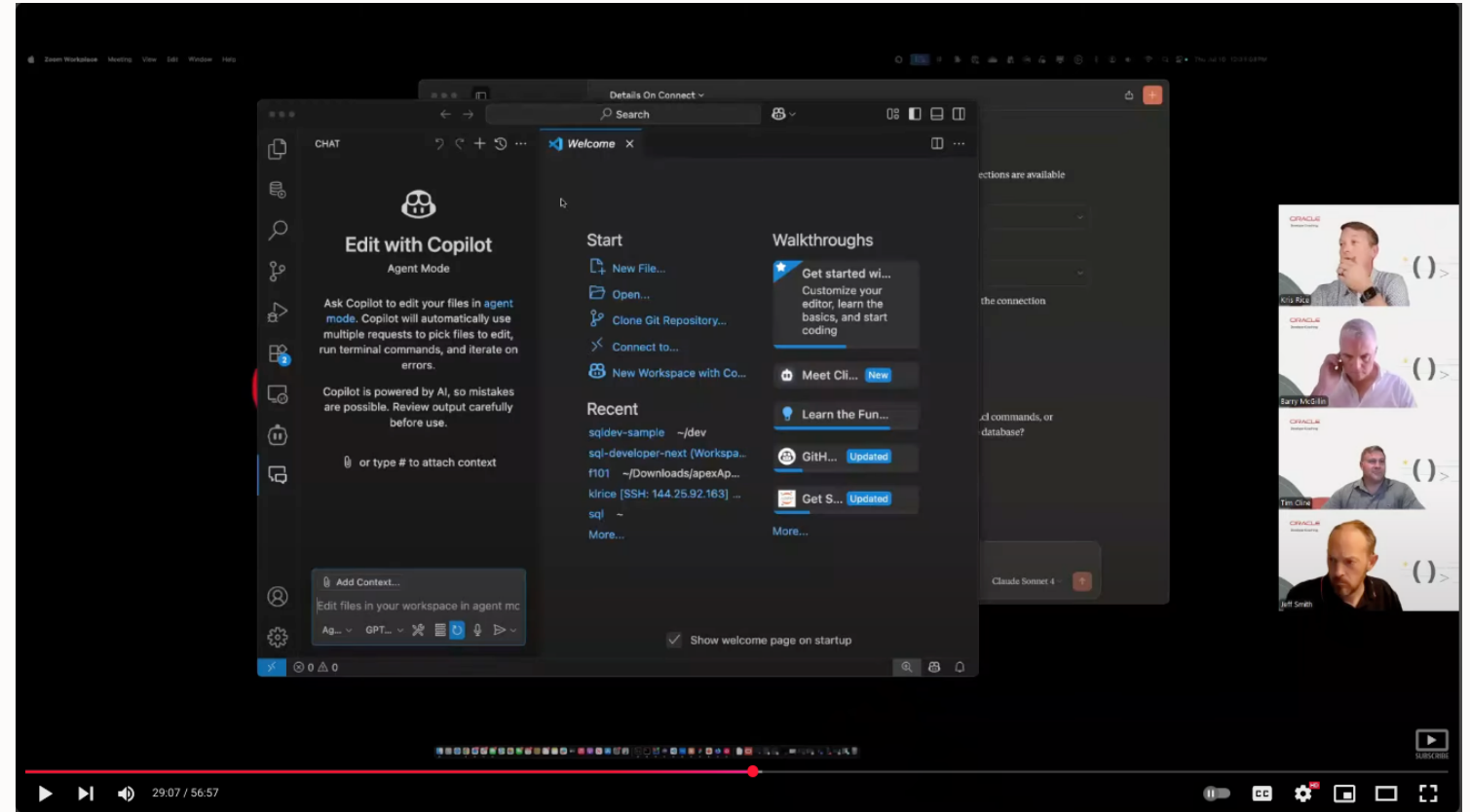
[Oracle SQLcl Download](#)

[Oracle MCP Server Blog Post](#)

[Oracle SQLcl MCP Documentation](#)

[Oracle SQL Developer for VS Code](#)

Coming Soon:
Hands-on LiveLab



**Your feedback is
important.**

**Scan this QR Code or use
the Mobile App to share
your thoughts on this
session.**



Thank you



No seriously...

Thank you.

ORACLE